BRHANE GIDAY

Portfolio: https://brhane.vercel.app
LinkedIn: https://linkedin.com/in/brhane-giday
GitHub: https://github.com/brhanegiday

Summary

Software & AI Engineer with over 6 years of experience building scalable full-stack applications, robust backend microservices, RESTful APIs, and secure workflow automation platforms across Africa. Proven expertise in Node.js, TypeScript, Python, PostgreSQL, Docker, and Kubernetes, with a strong focus on Digital Public Infrastructure (DPI) and Digital Public Goods (DPGs). Demonstrated track record in architecting high-availability end-to-end systems, integrating open-source frameworks (MOSIP), strengthening platform security, and translating complex public sector requirements into resilient, production-ready software solutions.

Experience

Software & AI Engineer Jan 2025 to Present
Upanzi Network - Kigali, Rwanda

- Architected and deployed high-performance Node.js and Python backend microservices and RESTful APIs, enabling seamless interoperability between legacy Digital Public Infrastructure (DPI) and open-source MOSIP systems.
- Strengthened platform security and trust boundaries by implementing OpenID Connect authentication, robust isolation practices, and secure credential handling across distributed services.
- Designed resilient CI/CD pipelines and containerized microservices using Docker and Kubernetes, ensuring high availability and fault tolerance across pan-African research networks.

Software Engineer Sep 2022 to Dec 2023
Better Mobile Security - Remote, USA

- Engineered secure backend APIs and data integration pipelines for mobile security applications, optimizing data flow and system reliability at scale.
- Collaborated across distributed international teams to enforce strict code quality, robust unit/integration testing standards, and backend security protocols.

Senior Frontend Engineer Mar 2022 to Sep 2022
DoubleDice - Remote, Belgium

- Architected scalable fullstack systems using Next.js, Node.js, and TypeScript with clean backend service abstractions and API integration patterns.
- Integrated Web3 wallet connectivity (MetaMask, WalletConnect) and smart contract interactions to enable seamless crypto transactions
- Led frontend QA, implementing unit tests and end-to-end testing with Jest

Frontend Developer & UI/UX Designer Sep 2020 to Oct 2021
MAD Technologies - Ethiopia

- Delivered client websites efficiently using WordPress and React projects.
- Designed intuitive user interfaces in Figma & Adobe XD.

Skills

- **Backend & Core Dev:** Node.js, RESTful APIs, Microservices Architecture, PostgreSQL, Python, System Design
- **Security & Identity:** DPI, OpenID Connect, OAuth2, Credentials & Key Management, Platform Security
- **DevOps & Infrastructure:** Docker, Kubernetes, Terraform, GitHub Actions, AWS (Certified CCP), Linux (CompTIA+)
- **DPG & Ecosystems:** Digital Public Infrastructure (DPI), Digital Public Goods (DPGs), MOSIP, Workflow Automation
- **AI & Data Engineering:** RAG Pipelines, Vector Databases, OpenAI APIs, FastAPI, Data Pipelines, LLM Orchestration

Education

MSc, Information Technology (Software Engineering & AI) Aug 2023 - May 2025
Carnegie Mellon University Africa, Rwanda

I focused on software engineering, DevOps, and AI with an emphasis on software design, architecture, and scalable systems. My coursework and projects explored distributed systems, cloud-native development, and applied machine learning, combining both research and hands-on practice.

BSc, Electrical & Computer Engineering Sep 2014- Jul 2019
Addis Ababa University, Ethiopia

I specialize in Computer Engineering, with a strong foundation in computer programming, computer architecture, digital logic design, and compilers.

Projects

MOSIP-Integrated Event Management System Backend & Integration Engineer

- Integrated MOSIP open-source identity verification APIs, enabling secure backend biometric and demographic token verification for event authentication.
- Designed secure role-based access control (RBAC) microservices and API endpoints for automated booking and event administration.
- Built a responsive dashboard using Angular Material and NgRx for state management.

OLGA- AI-Powered Academic Advisor Founder & Fullstack Engineer

- Architected and built full-stack AI-powered platform to help students plan academic paths aligned with career goals, integrating responsive client interfaces with secure backend services.
- Implemented CI/CD pipelines, wrote tests, and conducted code reviews.
- Collaborated with advisors and faculty, implementing workflows that streamline student–advisor interactions.

Fayda Chatbot System Backend & AI Systems Engineer

- Architected multi-channel webhook handlers and messaging worker services supporting Telegram and WhatsApp for automated National ID inquiries.
- Implemented support for multiple writing styles, including native Ge'ez script and phonetic Latin-script queries.

- Engineered a grounded RAG data pipeline and asynchronous human-in-the-loop review workflow service for National ID Agency staff verification.

Policy Analyzer Software & AI Engineer

- Engineered an automated policy analysis backend service using RAG pipelines to process unstructured regulatory texts for Digital ID and Data Protection frameworks across African nations.
- Designed interactive visual dashboards for international development stakeholders, enabling data-driven evaluations of Digital Public Infrastructure (DPI) readiness across target regions.

Nabayt- Storytelling Platform Fullstack Engineer

- Architected and developed an end-to-end full-stack platform for property rentals, connecting interactive frontend user interfaces with RESTful APIs and databases.
- Engineered secure, token-based authentication flows.
- Established robust CI/CD pipelines using GitHub Actions and Docker to automate testing and deployment.

KaziData – Labor Market Intelligence platform Fullstack & Data Engineer

- Architected full-stack labor-market data pipelines and backend microservices alongside interactive frontend interfaces to process job postings, curricula, and official labor data.
- Engineered a connected skills graph mapping engine and frontend analytics dashboards for workforce intelligence, skill gap analysis, and program alignment.
- Implemented multi-language localization supporting English, French, and Kinyarwanda across data integration workflows and user interfaces.

Certifications

- CompTIA Linux+, 2025-01-01
- AWS Certified Cloud Practitioner, 2025-01-10